Fiscal Year:	FY 2023	**Task Last Updated:**	FY 04/24/2024
PI Name:	Porada, Christopher Ph.D.		
Project Title:	Long-lived Single- and Multi-organ Tissue Equivalent (OTE) Platforms to Model the Response of Human Tissues to Various Stressors		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:	NIH, BARDA, FDA	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	27157	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA015N. Extended Longevity of 3D Tissues and Microphysiological Systems for Modeling of Acute and Chronic Exposures to Stressors
Start Date:	09/29/2022	**End Date:**	09/29/2026
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA HQ
Contact Monitor:	Koniges, Ursula	**Contact Phone:**	202-256-8786
Contact Email:	ursula.m.koniges@nasa.gov		
Flight Program:			
Flight Assignment:	Per the Research Announcement for this solicitation: this effort is a collaboration between NASA; the National Institutes of Health (NIH); the Biomedical Advanced Research and Development Authority (BARDA); and the Food and Drug Administration (FDA).		
Key Personnel Changes/Previous PI:	Jim Rice, PhD has been added as a senior research scientist in lieu of the post-doctoral fellows originally listed in the proposal. After searching for 6 months for qualified post-docs, Dr. Rice applied and we decided it was best to hire him instead of hiring the proposed 2 post-docs, as we did not receive any qualified applicants at the level of post-doc. As Dr. Rice already had extensive experience with organoids, he was a perfect fit. He started working on the project officially on June 1st of 2023		

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Grant/Contract No.:	80ARC022CA008
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Task Description:	Our human “body-on-chip” (hBoC) platform is comprised of organ tissue equivalents (OTEs) representing all major tissues in the body, connected in series to enable tissue-to-tissue communication. These OTEs are derived from normal human cells and accurately mimic human physiology, because each OTE is comprised of all major cell types present in its corresponding native tissue, in physiologic proportions (including innate immune components). For example, the liver OTE is 78% hepatocytes, 2% biliary cells, 5% stellate cells, 5% endothelial cells, and 10% Kupffer cells. The OTEs are designed to mimic each tissue’s natural 3-D structure, allowing appropriate cell-cell contacts, and they are supported within a microfluidic system that can be designed to house either a single OTE type or multiple OTE types integrated in series. The OTEs are embedded within a hydrogel that contains tissue-specific extracellular matrix (ECM) proteins and chemistry for tuning the mechanical stiffness to match that of the tissue being modeled. The physiological fidelity of the hBoC platform has been validated in pharmacology and toxicology studies. The liver OTE demonstrates acetaminophen induced toxicity that is reversed by its clinical countermeasure N-acetyl cysteine. Cardiac OTEs lose responsiveness to epinephrine in the presence of a #-blocker, and this response is recovered as the #-blocker is metabolized by an integrated liver OTE, underscoring the importance of integrated multi-organ systems. To date, our group has > 40 publications on the use of OTEs to: 1) demonstrate toxicities of recalled FDA-approved drugs; 2) better define the effects of environmental toxins and chemical warfare agents; and 3) model disease, such as infections by pathogenic organisms, including SARS-CoV-2. Individual OTE systems have also been exposed to radiation. Currently, our cardiac OTE maintains viability and function for > 6 mos., while the skin, liver, and lung OTEs have useable lifespans of > 5, 1, and 1.5 mos., respectively. Our overall goals are to extend the longevity of the skin, liver, and lung OTEs to 6 mos. and define the response and recovery of the 4 individual OTEs and the integrated hBoC after exposure to various stressors. We propose: Phase 1/Aim 1 (mos. 1-36): To extend the skin, lung, and liver OTEs’ usable lifespan to 6 mos. to match the cardiac OTE, we will employ a multi-pronged approach, testing the impact of: a) cell source; b) mechanical deformation and perfusion rate of the OTEs; and c) reducing media-to-tissue volume ratio to enable supportive paracrine signals to reach bioactive concentrations. The 4 single OTEs will be exposed to 3 types of stressors: a chemical toxin, an infectious agent, and ionizing radiation. We will use our developed on-board electrochemical and imaging sensors, which allow non-destructive monitoring, probing, and sampling for soluble biomarkers of viability, function, and overall health throughout the experimental timeframe. Phase 2/Aim 2 (mos. 37-48): The 4 OTEs will be integrated into one system and exposed to the stressor types of Aim 1. We will monitor exposure, adaptation, and recovery of tissues to the proposed perturbations with our on-board sensing system. Multi-omics and bioinformatics analyses will be performed on individual and integrated OTEs at predetermined timepoints to identify molecular pathways and potential novel biomarkers associated with both stressor-induced pathologic states and system recovery that may aid with the identification of countermeasures. Particular attention will be paid to initiation and cessation of reactive oxygen species (ROS), Senescence-associated secretory phenotype (SASP), and inflammation mediated by the innate immune cells in each OTE. Importantly, the liver OTEs secrete a full complement of serum factors, enabling modeling of protein binding and basic pharmacokinetics in drug and chemical toxin studies. All OTEs will be manufactured using cells from male and female donors of various ethnicities to address population diversity/heterogeneity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed studies will extend the usable lifespan of a highly translational, all-human organ tissue equivalent (OTE) platform to 6 months and validate its use as a model system for defining the risks to human tissues as a result of exposure to multiple spaceflight-relevant stressors.
	During the first year of the project, we performed a good deal of work on the design of the OTE “chips”, and we have now (after several months of work with both software-based simulations and prototype manufacturing and testing) succeeded in designing and creating a modular format that utilizes 3D printed nozzles to connect the fluidics system to the OTE “chip”, rather than tubing and epoxy, as we had been using at the time of grant submission. This advancement will greatly facilitate the transport of the OTE chips to the gamma irradiator to test the response of each human tissue to IR, once we have achieved a 6-month lifespan, and it represents a major stride in achieving chip-to-chip consistency and reproducibility. Since the submission of the grant, WFIRM has hired a new faculty member who is an expert in cardiac physiology and in the fabrication and use of cardiac OTEs to study cardiac biology, physiology, and response to stressors/ insults (Dr. Josh Maxwell). He has graciously agreed to contribute his expertise to this project in a collaborative fashion. Under Dr. Maxwell’s leadership, we have now succeeded in making fully functional cardiac organoids from human iPS cells, enabling us to make cardiac organoids in which all cell types are from the same human donor, thus eliminating any immune issues with the use of organoids in which different cell types are derived from different donors. One of the stated goals of the project is to integrate all 4 OTEs together in a single microfluidic device. During the first year of funding, WFIRM’s ongoing “Universal Media” Program has succeeded in formulating a media that is equally effective at supporting the growth of all 4 OTEs to be studied in this project (lung, liver, skin, and cardiac), which is a major step forward towards our goal of ultimately integrating all 4 OTEs into a single device to be able to compare and contrast each OTE’s response to the various stressors individually and in the presence of the other OTEs.

Task Progress:	During the first year of funding, the lung team succeeded in extending the usable lifespan of the lung OTE to 6 months, and the skin team has also succeeded in maintaining the skin OTE for 6 months. The liver has proven more difficult, and the liver team is concerned with achieving a 6-month lifespan with the liver OTE, given its complexity and the fragility of some of its cellular constituents. However, we had conversations with Lena Biosciences, who manufacturers the "PerfusionPal". This system uses unique artificial "blood" to enable parallel perfusion of 12 or 48 cultures, each of which can be in its own unique media and/or be exposed to its own stressor(s). This unique platform thus allows each organoid to be perfused "in-well", so that the concentrated secretome and metabolome remain in the well and can be assayed. The organoid in each individual well, exposed to each specific stressor, can also be collected for transcriptomic and proteomic analyses. Importantly, this system permits gas exchange at physiologic rates, which should greatly prolong the lifespan of organoids being cultured. We purchased this system and have begun to work with it to see whether it will enable us to achieve longer lifespan of the liver organoids, which are the only organoid that have not yet reached the targeted 6-month lifespan. We have also transitioned to using iPSC-derived cells to fabricate the liver OTEs in the hope that the combined use of iPSC-derived cells and culture in the PerfusionPal system will enable to reach our ambitious goal of 6-month longevity. Importantly, the use of iPS-derived cells will also allow us to fabricate liver OTEs in which all 4 cell types are derived from the same donor, thereby eliminating any confounding factors due to immunologic incompatibility between the donors from which the primary liver cells are derived. Our team members at Harvard's Brigham and Women's Hospital also made exciting progress towards our goals of being able to non-invasively monitor the OTEs in real-time over this first year of funding. During the first year, we also established a critical new collaboration with Dr. Eain Murphy, a faculty member at Upstate Medical University in Syracuse who brings decades of experience with herpes viruses – CMV in particular. Dr. Murphy has agreed to provide us with an entire family of fluorescent endotheliotropic CMV variants he has created that not only allow tracking of viral infection, latency, and reactivation, but also make it possible to track, in near real-time, the kinetics of CMV replication. These novel CMV variants and Dr. Murphy's expertise in CMV biology will be invaluable for the studies we proposed to examine CMV in the endothelium of human organoids. After nearly 6 months of advertising and reviewing applications, we hired Dr. Jim Rice as a Senior Research Scientist, in lieu of the 2 post-doctoral fellow positions we had originally planned. Dr. Rice brings extensive experience with human organoids, having already done 6 years of post-doctoral fellowships and co-founding a Biotech start-up, the focus of which was to generate human iPSC-derived brain organoids that would liver and function for at least a year and recapitulate multiple regions of the human brain. The start-up succeeded in achieving these ambitious goals, giving Dr. Rice a great deal of experience that will be invaluable for the current NASA project. We also commenced a collaboration with Egle Cekanaviciute and Piyumi Wjeeskara at NASA Ames, and with Bruce Demple at SUNY Stony Brook, who has one of the only publications assessing the effects of regolith simulants on human cells. He provided us with a wealth of valuable information, informing us that the regolith simulants need to be "activated" by grinding in mortar and pestle, and that the mortar and pestle must be made of pure agate to ensure no contaminants are released during the grinding that could impact the properties of the regolith. We also found an abstract from a group at NASA demonstrating that exposing regolith simulants to UV light is also critical to more accurately model the properties of native regolith, as the surface of the Moon is continually bombarded with UV radiation due to its lack of an atmosphere. We have now performed several runs exposing the human lung OTEs (spheroids and 96-well planar OTEs) to simulated lunar regolith, and we have observed that regolith exerts multiple effects on this human tissue. The skin team also made a great deal of progress testing the ability of the skin OTEs to serve as a much-needed in vitro platform for studying the response of human skin to acute and chronic UV irradiation, laying the necessary groundwork for our proposed studies with ionizing radiation. Dr. Atala bought a VITROCELL® PowderX Dry Powder Exposure System for an ongoing BARDA program at WFIRM and we performed our first exposures of the air-liquid-interface (ALI) lung/airway OTEs to regolith using this highly physiologically accurate exposure system. Data are currently being analyzed from this first-ever study delivering regolith to human lung/airway OTEs as a dry powder, just as will occur when astronauts inhale regolith. Dr. Rice also perfected an optical clearing technique that now enables us to image through the entire planar lung/airway OTE on the confocal and then flatten the resultant z-stack into a single image. Applying this technique allowed us to perform staining for ??-H2AX and demonstrate that, in this highly accurate inhalation model of exposure to regolith, there is significant DSB induction in the planar lung/airway OTEs as a result of regolith exposure.
Bibliography Type:	Description: (Last Updated: 08/05/2025)